$$24xy^3z = 2\times2\times2\times3\times x\times y^3\times z = 2^3\times3\times x\times y^3\times z = 2^3\times3\times x\times y\times y^2\times z$$

Therefore, $GCD = 2^3xy^2$

(ii) $y^3+1 = y^3+1^3 = (y+1)(y^2-y+1)$

$y^2-1 = y^2-1^2 = (y+1)(y-1)$

Therefore, $GCD = (y+1)$

(iii) $2x^2-18 = 2(x^2-9) = 2(x^2-3^2) = 2(x+3)(x-3)$

$x^2-2x-3 = x^2-3x+x-3$

$= x(x-3)+1(x-3)$

$= (x-3)(x+1)$

Therefore, $GCD = (x-3)$

(iv) $(a-b)^2, (b-c)^3, (c-a)^4$

There is no common factor other than one.

Therefore, $GCD = 1$

Exercise 3.9

1. Find the GCD for the following:

(i) p^5, p^{11}, p^9

(ii) $4x^3, y^3, z^3$

(iii) $9a^2b^2c^3, 15a^3b^2c^4$

(iv) $64x^8, 240x^6$

(v) $ab^2c^3, a^2b^3c, a^3bc^2$

(vi) $35x^5y^3z^4, 49x^2yz^3, 14xy^2z^2$

(vii) $25ab^3c, 100a^2bc, 125ab$

(viii) $3abc, 5xyz, 7pqr$

2. Find the GCD of the following:

(i) $(2x+5), (5x+2)$

(ii) $a^{m+1}, a^{m+2}, a^{m+3}$

(iii) $2a^2+a, 4a^2-1$

(iv) $3a^2, 5b^3, 7c^4$

(v) x^4-1, x^2-1

(vi) $a^3-9ax^2, (a-3x)^2$

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